ASSIGNMENT-1

Reg No: 24125096
Course code: MLA0205
Course Name: Fundamentals of machine learning

① Design of learning systems

case	learning Paradigm	learning task.
a) Supervised learning	classification	supervised learning
b) Customer segments	unsupervised learning	clustering
c) Credit-card spend	supervised learning	Regression
d) ATM cash refill	Reinforcement learning	control / decision Making.
e) fraud detection	Semi-Supervised learning	classification.

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② Loan Default:-
*T: Predict loan default
*E: 10 years of labelled relayment records.
*P: Accuracy, Precision and recall.

③ Customer Segmentation:-
*T. Group customers based on behaviour.
*E: 2 million unlabelled customer records.
*P: clustering quality.

④ credict-card spend:-
*T: Predict next-month spending
*E: historical customer spending records.
*P: Mean squared Error (MSE).

⑤ ATM cash refill:-
*T: learn the best cash-refill policy.
*E: Repeated trails and feedback
*P: Cash availability minus transfer cost.

⑥ fraud detection:-
*T: Identify fraudent transactions.
*E: 2-1. labelled and remaining unlabelled records.
*P: Precision, recall and F1-score.

(2) Baye's theorem:-

Given *Prevalence = 2% = 0.02

* Sensitivity = 95% = 0.95

* Specificity = 90% = 0.90

* False Positive rate = 0.10

using Baye's theorem

$$P(D|+) = \frac{P(+|D)P(D)}{P(+|D)P(D) + P(+|\bar{D})P(\bar{D})}$$

$$= \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 \times 0.98)}$$

$$= \frac{0.019}{0.019 + 0.098} = \frac{0.019}{0.117}$$

$P(D|+) = 0.1624 = 16.24\%$

Conclusion:- The Probability is lower than sensitivity because the disease prevalence is very low resulting in many false-Positive results.

(3) Joint Probability Distribution:-

X	Y=0	Y=1
0	0.10	0.20
1	0.15	0.25
2	0.20	0.10

(a) Marginal Probabilites:-

For X:

$P(X=0) = 0.10 + 0.20 = 0.30$

$P(X=1) = 0.15 + 0.25 = 0.40$

$P(X=2) = 0.20 + 0.10 = 0.30$

for Y:

$P(Y=0) = 0.10 + 0.15 + 0.20 = 0.45$

$P(Y=1) = 0.20 + 0.25 + 0.10 = 0.55$

$E(X) = \sum x P(x)$

$= 0(0.3) + 1(0.4) + 2(0.3)$

$\boxed{E(X) = 1}$

$E(X^2) = 0 + 1(0.4) + 4(0.3) = 1.6$

$\text{Var}(X) = E(X^2) - [E(X)]^2$

$= 1.6 - 1^2$

$\boxed{\text{Var}(X) = 0.6}$

$E(Y) = 0(0.45) + 1(0.55)$

$\boxed{E(Y) = 0.55}$

(c) Conditional Probability:-

$P(X = 2|Y = 1) = \frac{P(X=2,\ Y=1)}{P(Y=1)} = \frac{0.10}{0.55}$

$\boxed{0.1818}$

(d) Covariance and Correlation:-

$E(XY) = 0 + 0.25 + 0.20 = 0.45$

$\text{cov}(X,Y) = E(XY) - E(X)E(Y)$

$= 0.45 - (1)(0.55)$

$\boxed{\text{Cov}(X,Y) = -0.10}$

$\text{Var}(Y) = 0.55(0.45) = 0.2475$

$r = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)\ \text{Var}(Y)}}$

$= \frac{-0.10}{\sqrt{\text{Var}(X)\ \text{Var}(Y)}}$

$= \frac{-0.10}{\sqrt{0.6 \times 0.2475}}$

$\boxed{r = -0.2593}$

④ Normal Distribution and Maximum likelihood:

Given $X \sim N(50, 16)$

$\mu = 50, \quad \sigma = \sqrt{16} = 4.$

ⓐ Calculate Probabilities

1. Find $P(X > 58)$

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{58 - 50}{4} = 2$$

$$P(X > 58) = 1 - \Phi(2)$$

$$= 1 - 0.9772$$

$$\boxed{P(X > 58) = 0.0228}$$

2. Find $P(46 < X < 54)$

$$Z_1 = \frac{46 - 50}{4} = -1$$

$$Z_2 = \frac{54 - 50}{4} = 1$$

$$P(-1 < Z < 1) = \Phi(1) - \Phi(-1)$$

$$= 0.8413 - 0.1587$$

$$\boxed{P(46 < X < 54) = 0.6826}$$

ⓑ Maximum likelihood

Given readings:

12, 15, 11, 14, 13

Mean:

$$\hat{\mu} = \frac{12 + 15 + 11 + 14 + 13}{5}$$

$$\boxed{\hat{\mu} = 13}$$

variance:

$$\hat{\sigma}^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

$$= \frac{(-1)^2 + 2^2 + (-2)^2 + 1^2 + 0^2}{5}$$

$$= \frac{10}{5}$$

ML variance = 2

unbaised variance:

$$s^2 = \frac{10}{5 - 1}$$

unbaised variance = 2.5

Conclusion:- ML variance is baised because it divides by n, where as unbaised variance divides by n-1.

⑤ Information Theory - Entropy

Given Probabilities

$$P = (1/2, 1/4, 1/8, 1/8)$$

ⓐ calculate entropy:-

$$H(X) = -\sum P(x) \log_2 P(x)$$

$$= -\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$\boxed{H(X) = 1.75 \text{ bits}}$$

niform source entropy:-

$H(x) = \log_2 4$

$H(x) = 2$ bits

Redundancy:

$R = 2 - 1.75$

$R = 0.25$ bits

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Conclusion:- the code is optimal because its average length equals entropy.

(c) Average code length:-

Given code: 111

Code lengths:- 1, 2, 3, 3.

$L = \sum P(x_i) l_i$

$= \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$

$L = 1.75$ bits

(b) KL Divergence

Given:-

$P = (0.5, 0.25, 0.25)$

$Q = (1/3, 1/3, 1/3)$.

(a) calculate $KL(P||Q)$

$D_{KL}(P||Q) = \sum P(x) \log_2 \frac{P(x)}{Q(x)}$

$= 0.5 \log_2(1.5) + 0.25 \log_2(0.75) + 0.25 \log_2(0.75)$

$D_{KL}(P||Q) = 0.085$ bits

(b) calculate $KL(Q||P)$:-

$D_{KL}(Q||P) = \sum Q(x) \log_2 \frac{Q(x)}{P(x)}$

$= \frac{1}{3} \log_2 \frac{1/3}{0.5} + \frac{1}{3} \log_2 \frac{1/3}{0.25} + \frac{1}{3} \log_2 \frac{1/3}{0.25}$

$D_{KL}(Q||P) = 0.0817$ bits

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Conclusion:-

KL divergence is asymmetric because:

$D_{KL}(P||Q) \neq D_{KL}(Q||P)$

Therefore, changing the order of distributions changes the training loss.

(7) Entropy and Mutual Information:-

Given:

$P(0,0) = 0.4$, $P(0,1) = 0.1$

$P(1,0) = 0.1$, $P(1,1) = 0.4$

(a) Calculate H(A) and H(B):-

Marginal Probabilities:

$P(A=0) = 0.5$, $P(A=1) = 0.5$

$P(B=0) = 0.5$, $P(B=1) = 0.5$

$H(A) = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5)$

$H(A) = 1$ bit

Similarly,

$H(B) = 1$ bit

(b) Joint entropy:-

$H(A,B) = -\sum P(a,b) \log_2 P(a,b)$

$= -[2(0.4 \log_2 0.4) + 2(0.1 \log_2 0.1)]$

$H(A,B) = 1.722$ bits

(c) Conditional entropy:-

$H(A|B) = H(A,B) - H(B)$

$= 1.722 - 1$

$H(A|B) = 0.722$ bits

(d) Mutual information:-

$I(A;B) = H(A) - H(A|B)$

$= 1 - 0.722$

$I(A;B) = 0.278$ bits

Conclusion:- B Provides some information about A and can be used as a useful feature for Predicting A, but it does not completely determine A.

(8) Bayes-Decision theory:-

$P(x|\omega_1) = N(2,4)$

$P(x|\omega_2) = N(6,4)$

$P(\omega_1) = 0.6$, $P(\omega_2) = 0.4$

(a) Derive Bayes decision threshold:-

for equal variance and zero-one loss:

$P(\omega_1) P(x|\omega_1) = P(\omega_2) P(x|\omega_2)$

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_2 - \mu_1} \ln \frac{P(\omega_1)}{P(\omega_2)}$$

$$= \frac{2+6}{2} + \frac{4}{6-2} \ln \frac{0.6}{0.4}$$

$= 4 + \ln(1.5)$

$x^* = 4.4055$

assify $x = 4.5$

$4.5 > 4.4055$

$x = 4.5$ belongs to ω_2

(c) Overall Probability of error :-

$P(\text{error}) = P(\omega_1) P(x > x^* | \omega_1) + P(\omega_2) P(x < x^* | \omega_2)$

$= 0.6 [1 - \Phi(1.2029)] + 0.4 [\Phi(-0.7731)]$

using standard normal values.

$\approx 0.6 (0.1146) + 0.4 (0.2194)$

$= 0.06876 + 0.08776$

$P(\text{error}) \approx 0.1564$ 15.36%

(d) Effect of increasing Prior :-

If the Prior Probability of ω_1 increases, the threshold moves towards ω_2. This increases the decision region for ω_1.

(9) Minimum Risk Decision :-

Given:

$P(\omega_1|x) = 0.3, \quad P(\omega_2|x) = 0.7$

(a) Risk of declaring disease :-

$R(\alpha_1) = 0(0.3) + 8(0.7)$

$R(\alpha_1) = 5.6$

(b) Risk of declaring healthy :-

$R(\alpha_2) = 50(0.3) + 0(0.7)$

$R(\alpha_2) = 15$

(c) Minimum risk :-

$5.6 < 15$

Declare disease

(d) MAP decision :- MAP declares healthy because 0.7 is greater than 0.3. Minimum-risk decision declares disease.

(10) Bias-variance trade off:

given:

Degree	1	3	5	7
Training MSE	9.0	4.1	2.2	6.9	8.1
validation MSE	9.5	4.6	2.9	5.6	11.8

(a) Generalisation gap:-

Gap: validation MSE - Training MSE

Degree	Gap
1	0.5
2	0.5
3	0.7
5	2.7
7	11.7

(b) Model diagonsis:-

* Degree 1 and 2: underfitting.
* Degree 3: Good balance.
* Degree 5: Overfitting.
* Degree 9: servere overfitting.

(c) Best Model:-

Degree 3

Reason: lowest validation MSE = 2.9.

(d) Improvement techniques:-

1. Apply L1 or L2 regularisation.
2. Increase training data.

explain

Heads = 7, Tails = 3

Prior: Beta (2,2)

(b) MLE :-

$$\hat{\theta}_{ML} = \frac{7}{10} = \boxed{0.7}$$

(c) MAP :-

$$\hat{\theta}_{ML} = \frac{7}{10} = 0.7.$$

Beta (2+7, 2+3) = Beta (9,5)

$$\hat{\theta}_{MAP} = \frac{9-1}{9+5-2} = 0.6667.$$

(c) Posterior mean :-

$$E(\theta) = \frac{7}{14} = 0.6429$$

(d) For 70 heads in 100 tosses :-

Estimate	Value :-
MLE	0.7000
MAP	0.6961
Posterior Mean	0.6923

Conclusion :-

With more data, the influence of the Prior decreases.

Hospital Early - Warning system :-

(a) Design the learning system :-

- T: Predict sepsis within 6 hours.
- P: Sensitivity, Specificity and F1 Score.
- E: Historical Patient records.
- Target: Probability of sepsis.
- Input: vital signs and laboratory values.
- Algorithm: Logistic regression.
- Validation: Train-test split and Cross-Validation.

TN = 950 (0.8) = 760

FP = 950 − 760 = 190

(c) Calculate PPV and NPV:-

$$PPV = \frac{TP}{TP+FP} = \frac{45}{45+190}$$

PPV = 19.15%

$$NPV = \frac{TN}{TN+FN} = \frac{760}{760+5}$$

NPV = 99.35%

(d) Calculate total cost:-

Missed Sepsis cost = £100

False alarm cost = £5

(i) using the model

Cost = (FN × 100) + (FP × 5)

= (5 × 100) + (190 × 5)

= 500 + 950

Cost = £1450

(ii) Alerting every patient:-

Cost = 950 × 5

Cost = £4750

(iii) Never alerting

Cost = 50 × 100

Cost = £5000

Cost comparison:-

Method	Total cost
using model	£1450
Alerting everyone	£4750
Never alerting	£51000